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HS-OBS	Functional Requirements Specification
DEPARTMENT	DOCUMENT TYPE

VZI R1

Operating Modes Functional Requirements Specification
Toilet system

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00	11/09/2023	A. Santini	First Issue based on EC01P008888B Rev.00.6 VZI-50 Toilet system SFRS (parts removed in crossed red and parts added in blue).

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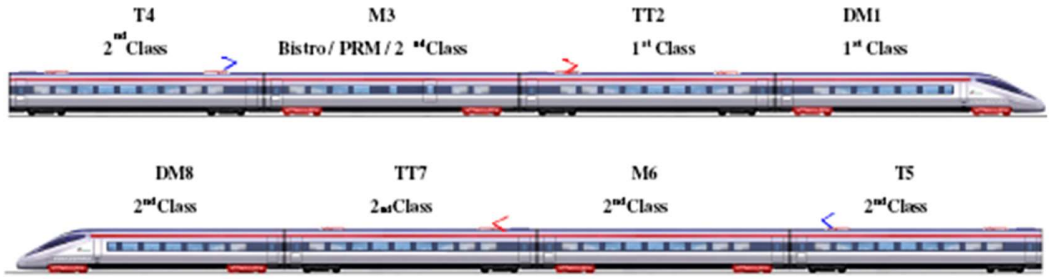
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1 Introduction

This document is a function specification for the control of the Toilet system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

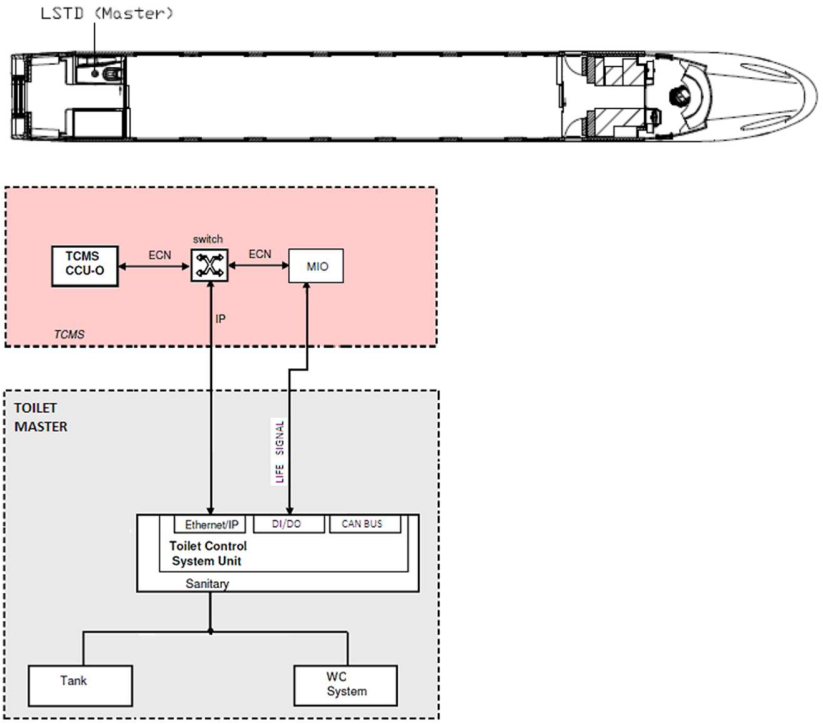
CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System
LSTD	Left hand standard toilet
RSTD	Right hand standard toilet
UNI	Universal toilet

2 System Overview

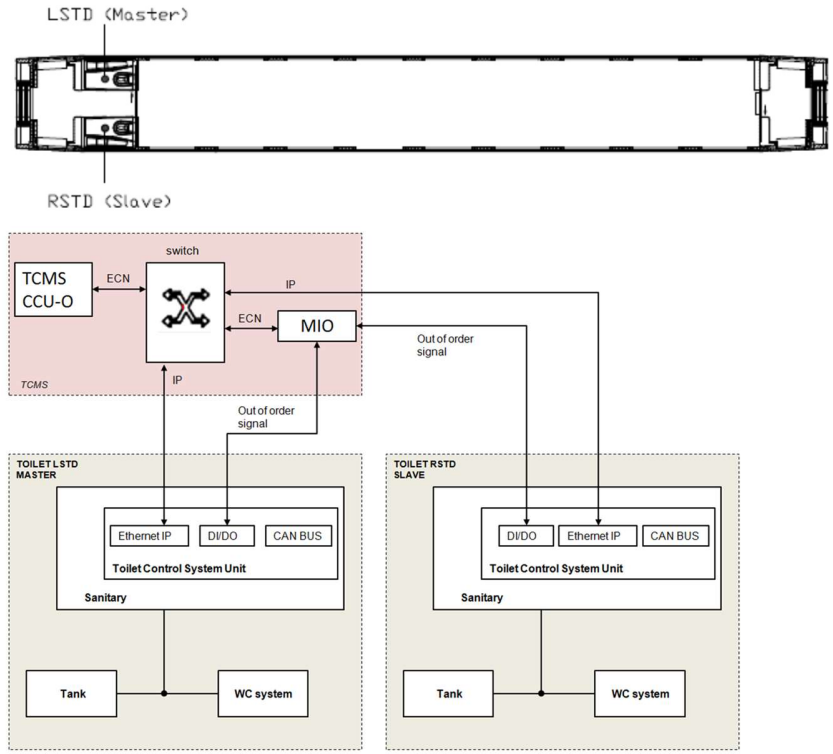
1.3 System Configuration

The toilet system is composed by four different kits:

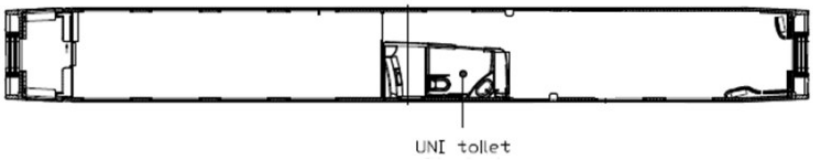
- DM KIT composed by 1 left standard toilet (LSTD);
- T KIT composed by 1 left standard toilet (LSTD) and 1 right standard toilet (RSTD);
- M3 KIT composed by 1 universal toilet (UNI) ~~and 1 toilet right standard toilet (RSTD) dedicated to the bistrot personnel~~;
- EXE KIT composed by 1 executive toilet (EXE).

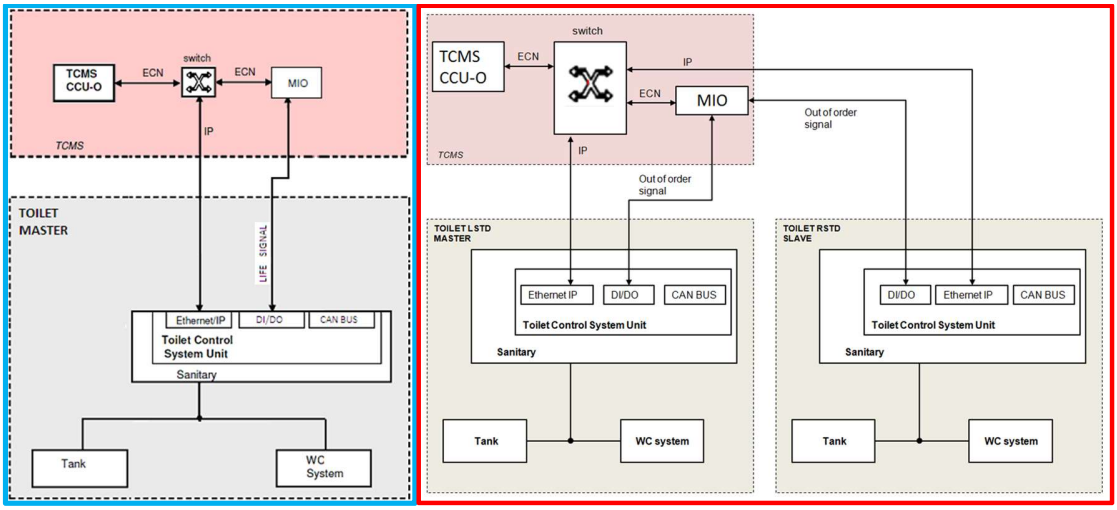


Picture 2 - *DM KIT configuration* (fitted on DM8)

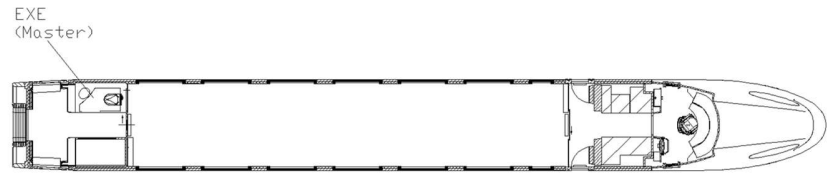


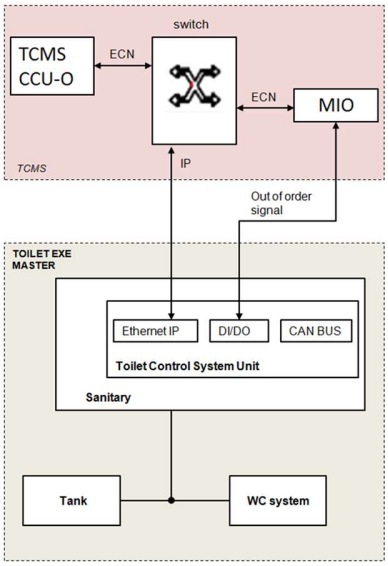
Picture 3 - T KIT configuration (fitted on TT2, T4, T5, M6 and TT7)





Picture 4 - M3 KIT configuration (M3 only)





Picture 5 - EXE KIT configuration (fitted on DM1)

3 Functional requirements

3.1 Toilet

3.1.1 Status of Toilets

3.1.1.1 Design description

The actual status of each toilet is identified by suitable devices installed both near the toilet itself (in the vestibule) and in the passenger compartment.

Vestibule indicators

The indications foreseen in the vestibule are completely handled by the toilet system without involving the train circuitry and TCMS. LEDs panel indicator will be fitted on the toilet lining (one LEDs panel for each toilet) providing information about the actual status of the system (vacant/occupied/out of order).

For the UNI toilets additional indications are provided in order to signal the SOS alarm activation (optic and acoustic signals).

Compartment indicators

The indications foreseen in the passenger compartments are completely handled by the toilet system without involving the TCMS. LEDs panel indicators (one for each toilet) will be fitted on the end compartment wall providing information about the actual status of the system (vacant/occupied or out of order).

The LEDs panel will be fitted only on the end compartment wall close to the toilet system.

3.1.1.1.1 Normal condition

See previous chapter.

3.1.1.1.2 Failure cases

a) Vestibule LEDs panel indicator failure. Loss of toilet status indication in the vestibule;
b) Passenger compartment LEDs panel indicator failure. Loss of toilet status indication in the passenger compartment.

3.1.1.1.3 Failure detection

a) By inspection during the periodic check done by the maintenance staff;
b) By inspection during the periodic check done by the maintenance staff.

3.1.1.1.4 Failure handling

a) No action for the driver and crew. Failed LEDs panel has to be changed during maintenance;
b) No action for the driver and crew. Failed LEDs panel has to be changed during maintenance.

3.1.1.2 Requirements

ID	2F_08.02.Zefiro-Europe.TRS.70
Description	Suitable devices, located outside the toilet, shall give the status: occupied, out of order or free for use.
Safety level	N/A

The indications foreseen in the vestibule are completely handled by the toilet system without involving the train circuitry and TCMS. LEDs indicators will be fitted on the toilet lining (for each toilet there will be one set of status indicator) providing information about the actual status of the system (vacant/occupied/out of order).

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.02.Zefiro-Europe.CONCEPT.49	The Toilet system shall show the status of each toilet (vacant/occupied/out of order), the status indicators shall be located outside the toilet in the vestibule.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A

ID	2F_08.02.Zefiro-Europe.TRS.71
Description	Suitable devices, located in the passenger compartment, shall show the current toilet status: occupied, out of order or free for use.
Safety level	N/A

The indications foreseen in the passenger compartments are completely handled by the toilet system without involving the TCMS. LEDs panel indicators (one for each toilet) will be fitted on the end compartment wall providing information about the actual status of the system (vacant/occupied/out of order).

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.02.Zefiro-Europe.CONCEPT.50	The Toilet system shall show the status of each toilet (vacant/occupied or out of order), the status indicators shall be located in the passenger compartment.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A

3.1.2 Universal Toilet for Disabled

3.1.2.1 Design description

The information “UNI toilet alarm activated” is shown on both driver cab IDUs and cabin crew IDU. Considering that the toilet system will work also in case of IP communication failure, the information “UNI toilet alarm activated” is sent to the TCMS by a dedicated hardware line.

3.1.2.1.1 Normal condition

When the “alarm” is activated in the UNI toilet an acoustic and visual signal is activated outside the toilet system. This is done by the toilet itself.
In addition to the above reaction, the alarm activation is sent to the driver/crew also.

3.1.2.1.2 Failure cases

Toilet circuitries failure. Loss of alarm signaling to the driver/crew IDU.

3.1.2.1.3 Failure detection

a) Failure not detectable automatically. It will be detect during the periodic checks performed by the maintenance staff.

3.1.2.1.4 Failure handling

a) The failed circuitries have to be changed during maintenance.

3.1.2.2 Requirements

ID	2F_08.02.Zefiro-Europe.TRS.76
Description	The activation of the alarm in the UNI toilet shall be signalled in the crew cabin and in the drivers cab.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.02.Zefiro-Europe.CONCEPT.58	The activation of the UNI toilet alarm shall be sent to the TCMS using a dedicated harwired line.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.107	The activation of the UNI toilet alarm shall be sent to the TCMS using a dedicated harwired line	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_08.02.Zefiro-Europe.CONCEPT.108	TCMS shall store an event (Warning) in case of activation of the UNI toilet alarm.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_08.02.Zefiro-Europe.CONCEPT.159	The toilet system shall indicate, outside the toilet, when the UNI toilet alarm is activated in both visual and acoustic way.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A

3.2 Tank/pipe control and supervision

3.2.1 Fresh Water supply supervision

3.2.1.1 Design description

The actual level of the FWT of each coach is sent to the TCMS via IP for supervision purpose. In the coach with two toilets this function is performed by the master one.
The FWT levels transmitted to the TCMS are 0%, 25%, 50%, 75% and 100%. They are shown on a suitable IDU page where the information coming from all coaches are represented.

3.2.1.1.1 Normal conditions

Continuous monitoring of the FWT level.

3.2.1.1.2 Failure cases

a) Level transducer failure. Loss of FWT levels indication on the cabin crew IDU.

3.2.1.1.3 Failure detection

a) Failure condition detected by the tanks level control unit.

3.2.1.1.4 Failure handling

a) No action for the driver/crew. Level transducer has to be changed during maintenance.

3.2.1.2 Requirements

ID	2F_08.02.Zefiro-Europe.TRS.26
Description	Information on the level in the fresh water tanks shall be monitored. Information about low level shall be shown on the IDU in the crew cabin.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.02.Zefiro-Europe.CONCEPT.51	Information about the actual level of the FWT (IFWTankContent) shall be sent to the TCMS.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.110	TCMS shall receive the actual level of the FWT (IFWTankContent) of each coach and show them on a suitable IDU page.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_08.02.Zefiro-Europe.CONCEPT.111	The FWT empty signal (IFWTankEmpty) shall be sent to the TCMS when the FWT is empty.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.112	TCMS shall store an event (Warning) in case of activation of FWT empty signal (IFWTankEmpty).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.2.2 Retention tank supervision

3.2.2.1 Design description

The actual level of the WWT of each coach is sent to the TCMS via IP for diagnostic and supervision purpose. In the coach with two toilets this function is performed by the master one.

The WWT levels transmitted to the TCMS are 50%, 80% and 95%.

3.2.2.1.1 Normal condition

Continuous monitoring of the WWT level.

3.2.2.1.2 Failure cases

a) Level transducer failure. Loss of WWT levels indication on the cabin crew IDU.

3.2.2.1.3 Failure detection

a) Failure condition detected by the tanks level control unit.

3.2.2.1.4 Failure handling

a) No action for the driver/crew. Level transducer has to be changed during maintenance.

3.2.2.2 Requirements

ID	2F_08.02.Zefiro-Europe.TRS.30
Description	Information on the level in the retention tank shall be monitored to determine if the toilet door needs to be locked to deny access to passengers. Information about high level shall be shown on the IDU in the crew cabin with an instruction to lock the toilet door.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.02.Zefiro-Europe.CONCEPT.52	Information about the actual level of the WWT shall be sent to the TCMS.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.114	TCMS shall receive the actual level of the WWTs of each coach and show them on a suitable IDU page.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_08.02.Zefiro-Europe.CONCEPT.115	TCMS shall store an event (Warning) when the WWT is completely full.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_08.02.Zefiro-Europe.CONCEPT.116	TCMS shall store an event (Warning) when the WWT is 80 % full.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.2.3 Heating of Water Pipes and Waste Water Tank

3.2.3.1 Design description

Heating of water pipes is not needed due to their routing (inside car).

Waste water tank heating is handled by the toilet system without involving the TCMS. Suitable heating elements fitted on the bottom side of the waste water tank are controlled by a thermostat that activates them when a temperature lower

than a preset value (tbd) is detected.
The heating elements will be fed by the medium voltage system (230Vac).

3.2.3.1.1 Normal condition

Suitable heating elements are controlled by a thermostat according to the current temperature of the waste water inside the WWT.

3.2.3.1.2 Failure cases

Heating element/thermostat failure. Loss of WWT heating function.

3.2.3.1.3 Failure detection

Failure condition detectable during the WWT empting operation (ice forming).

3.2.3.1.4 Failure handling

a) No action for the driver/crew. Heating element/thermostat has to be changed during maintenance.

3.2.3.2 Requirements

ID	2F_08.02.Zefiro-Europe.TRS.74
Description	Heating of waste water tanks shall be available to ensure that the waste water does not freeze.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.02.Zefiro-Europe.CONCEPT.68	In order to prevent freezing of waste water, the waste water tank shall be fitted with suitable heating devices.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.140	The heating devices shall be activated by a thermostat.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.141	The heating devices shall be fed by the medium voltage system (230 Vac).	Derived Requirement	4S_08.03.-.Toilet Systems		N/A

3.2.4 Complete Freeze Drainage

3.2.4.1 Design description

The complete freeze draining procedure empties the complete piping of the fresh water system including the fresh water tank. The complete freeze draining procedure is handled by the Master toilet cabin. The complete freeze drain procedure can be actuated by a signal via Ethernet or locally via the MMI interface of the toilet control unit. ~~or a thermostat (a dedicated thermostat for the total freeze drain function is included in toilet system) or a low voltage status~~.

In addition, it is possible to empty the fresh water tank acting on a manual ball valve.

~~The complete freeze drain procedure is available also in case of the toilet system is not powered (due to the operational mode of the train).~~

Complete freeze draining via Ethernet (train/toilet powered)

When the complete freeze draining procedure is activated via Ethernet (command issued by IDUs) the complete freeze drain valve opens and, after a preset time, the “partial freeze drain routine” starts.

When the partial freeze draining is finished the “freeze drain done” feedback is sent to the TCMS.

When the complete freeze drainage has been done the complete freeze drain valve will return to its initial state once the signal for freeze drain is switched off.

If the “partial freeze drain” procedure has already been done when the “complete freeze drain” command is received by the toilet system, only the complete freeze drain valve is opened (the partial freeze drain routine is not performed again).

Complete freeze draining via the MMI interface of the toilet control unit (train/toilet powered)

When the complete freeze draining procedure is activated via the MMI interface of the toilet control unit the complete freeze drain valve opens and, after a preset time, the “partial freeze drain routine” starts.

When the partial freeze draining is finished the “freeze drain done” feedback is sent to the TCMS.

When the complete freeze drainage has been done the complete freeze drain valve will return to its initial state once the command for freeze drain is deactivated via the MMI interface.

If the “partial freeze drain” procedure has already been done when the “complete freeze drain” it’s requested, only the complete freeze drain valve is opened (the partial freeze drain routine is not performed again).

~~**Complete freeze drain via thermostat (train/toilet powered)**~~

~~If a temperature below the preset value for the “complete freeze drain” procedure is detected, the complete freeze drain valve opens empting the fresh water tank. In this condition the “partial freeze drain” routine has already been performed due to the fact that the temperature threshold ($T_{cfd\downarrow}$) that commands the complete freeze drain routine is lower than the one used to start the partial freeze drain routine ($T_{cfd\downarrow} < T_{pfd\downarrow}$).~~

~~The complete freeze drain valve will close again if the temperature rises above a preset value ($T_{cfd\uparrow}$).~~

~~**Complete freeze drain via thermostat and low voltage status (train/toilet not powered)**~~

~~If the train is switched off (partial freeze draining procedure has already been done) a privileged voltage will be supplied from the train batteries to the thermostat control unit (included in the toilet) to keep it activated. In this condition only the fresh water tank will be emptied.~~

~~If a temperature below a preset value ($T_{cfd\downarrow}$) is detected by the thermostat control unit, the complete freeze drain valve opens empting the fresh water tank. The valve will close again if the temperature rises above the preset value ($T_{cfd\uparrow}$).~~

~~In case the batteries are low the thermostat control unit detects the low voltage status (TBD value) and the complete~~

~~freeze drain valve opens empting the fresh water tank. In this case the valves will remain open until the voltage is up again or the train is switched on and proper battery voltage is restored.~~

3.2.4.1.1 Normal condition

The complete freeze drainage is normally requested by the staff via IDU when the train has to remain unpowered for a period exceeding 12 hours in the presence of temperatures close to or below, zero or for example due to important maintenance activities. ~~for a long time (i.e. due to important maintenance activities).~~

~~The thermostat unit is used as a back up in case of the normal procedure is not performed correctly by the staff.~~

3.2.4.1.2 Failure cases

a) FWT empting Electro-valve failure. Loss of complete freeze drainage function.

3.2.4.1.3 Failure detection

a) Failure condition detected by the toilet control unit.

3.2.4.1.4 Failure handling

a) No action for the driver/crew. Empting Electro-valve has to be changed during maintenance.

3.2.4.2 Requirements

ID	2F_08.02.Zefiro-Europe.TRS.43
Description	In a situation where the temperature outside the train is low and it is a risk of freezing in the water tanks an automatic or manual drainage of the water tanks shall be activated
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
~~2F_08.02.Zefiro-Europe.CONCEPT.54~~	~~If a temperature below a preset value [Tcfd↓] is detected inside the toilet piping, the complete freeze drain valve shall open empting the fresh water tank.~~ ~~The system shall come back to the normal operation when a temperature above a preset value is detecetd [Tcfd↑].~~	~~Derived Requirement~~	~~4S_08.03.-.Toilet Systems~~		~~N/A~~
2F_08.02.Zefiro-Europe.CONCEPT.121	The toilet system shall activate the complete freeze drainage when it is ordered by TCMS.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.122	It shall be possible to set and reset the “complete freeze drain” command of all toilets from the IDUs.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

2F_08.02.Zefiro-Europe.CONCEPT.192	When the toilet system receives the complete freeze drain command from the TCMS, the complete freeze drain valve shall open and, after a preset time, the “partial freeze drain” routine shall be performed.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.193	The toilet system shall remain in the “complete freeze drain” mode until the “complete freeze drain” command coming from the TCMS is active.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.194	Once activated the “complete freeze drain” command shall be deactivated by the TCMS when the following conditions are verified: - all the “partial freeze drain done” feedback coming from each toilet are active - all the “fresh water tank empty” signals coming from each master toilet are active	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
~~2F_08.02.Zefiro-Europe.CONCEPT.195~~	~~The toilet system shall be able to perform the fresh water tank empting also in case the main power supply of the system is not present.~~	~~Derived Requirement~~	~~4S_08.03.-.Toilet Systems~~		~~N/A~~
~~2F_08.02.Zefiro-Europe.CONCEPT.196~~	~~For each fresh water tank, a low power consumption unit (thermostat control unit) able to monitor the temperature of the water inside the toilet piping shall be foreseen. That unit shall be fitted in the “master” toilet~~	~~Derived Requirement~~	~~4S_08.03.-.Toilet Systems~~		~~N/A~~
~~2F_08.02.Zefiro-Europe.CONCEPT.197~~	~~In addition to the main power supply line, a permanent power supply line (always present) shall be foreseen to the “master” toilet.~~	~~Derived Requirement~~	~~4S_09.02.09.-.Conventional Train Control - Circuits~~		~~N/A~~
~~2F_08.02.Zefiro-Europe.CONCEPT.198~~	~~The thermostat control unit shall be able to detect a low voltage status.~~	~~Derived Requirement~~	~~4S_08.03.-.Toilet Systems~~		~~N/A~~
~~2F_08.02.Zefiro-Europe.CONCEPT.199~~	~~If the thermostat control unit detects a low voltage status the complete freeze drain valve shall open empting the fresh water tank.~~ ~~The system shall come back to the normal operation when a proper battery voltage is restored.~~	~~Derived Requirement~~	~~4S_08.03.-.Toilet Systems~~		~~N/A~~
2F_08.02.Zefiro-Europe.CONCEPT.200	It shall be possible to empty the fresh water tank acting on a manual valve.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.HR01	The toilet system shall activate the complete freeze drainage when it is ordered via the MMI interface of the toilet control unit.	New Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.HR02	It shall be possible to set and reset the “complete freeze drain” command via the MMI interface of the toilet control unit.	New Requirement	4S_08.03.-.Toilet Systems		N/A

3.2.5 Partial Freeze Drainage

3.2.5.1 Design description

The partial freeze draining routine empties the complete piping of the fresh water system and the toilet itself.

The partial freeze drain routine can be actuated by a signal via Ethernet or locally via the MMI interface of the toilet control unit ~~or a thermostat (a dedicated thermostat for the partial freeze drain function is included in toilet system).~~

When the partial freeze drain routine is activated the partial freeze drain valve closes, the water tap is opened and the water heater and water panel are emptied. Parallel to this the toilet starts the emptying cycles until the level sensor in the flush water tank does not detect water any more. After this a preset number of final cycles will be performed.

When the partial freeze draining routine is completed the “partial freeze drain done” signal is sent by each toilet to the TCMS via Ethernet.

The partial freeze drain valve will remain closed after the freeze drain routine is completed until the “partial freeze drain” command (from Ethernet or via the MMI interface of the toilet ~~or from thermostat control unit~~) is present.
Air supply and electrical power supply are needed to perform partial freeze drain routine. Considering that, the partial freeze draining will be performed during the shutdown procedure of the train (i.e before opening the main battery contactor).
In order not to delay the shutdown procedure the driver can start the partial freeze drain routine manually at the end of service. TCMS will only start the partial freeze drain if it has not been done and if it is winter conditions.

Partial freeze draining via Ethernet (toilet powered)
The “partial freeze drain command” is generated by the TCMS, and sent to all the toilets via Ethernet, in the following cases:

- when an air pressure below a preset value in the main pipe is detected by the TCMS;
- when the shutdown procedure (i.e before opening the main battery contactor) of the train is requested;
- when the partial freeze draining routine is requested by the staff through the IDU.

The toilet system will remain in the “partial freeze drain mode” as long as the command from Ethernet is active.

Partial freeze draining via the MMI interface of the toilet control unit (toilet powered)
The “partial freeze drain command” is requested via the MMI interface of the toilet control unit.
The toilet system will remain in the “partial freeze drain mode” as long as the command is deactivated via the MMI interface.

~~**Partial freeze drain via thermostat (toilet powered)**~~
~~A dedicated thermostat, fitted inside the toilet piping, monitors the temperature of the water.~~
~~If a temperature below a preset value ($T_{pfd↓}$)is detected by the thermostat control unit the partial freeze draining routine starts.~~
~~The routine will be turned off when the temperature rises above a preset value ($T_{pfd↑}$).~~

3.2.5.1.1 Normal condition

The partial freeze drainage is normally requested by the TCMS during the shutdown procedure of the train or in case of a low pressure value of the pneumatic system is detected.

3.2.5.1.2 Failure cases

a) Partial freeze Electro-valve failure (the valve is not able to interrupt the water feed). Loss of partial freeze drainage function.

3.2.5.1.3 Failure detection

a) Failure detected by the toilet control unit.

3.2.5.1.4 Failure handling

a) No action for the driver/crew. Empting Electro-valve has to be changed during maintenance.

3.2.5.2 Requirements

ID	2F_08.02.Zefiro-Europe.TRS.72
Description	If the air pressure is too low the water shall be drained If the air pressure is too low the water that is held in the fresh water tubing piping must be drained
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.02.Zefiro-Europe.CONCEPT.55	When an air pressure below 7 bar in the main pipe is detected and the minimum value in the last 24 hours of outside air temperature was below 5 °C for at least 1 hour, the TCMS shall set the “partial freeze drain” command, if not already requested by IDU, and send it to the toilet system via Ethernet. This command shall be set only when the following condition are fulfilled: - No driver deck enabled - No auxiliary voltage is present - Passenger doors not released The “low temperature time” evaluation is reset at each TCMS start up.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_08.02.Zefiro-Europe.CONCEPT.210	The TCMS shall reset the “partial freeze drain” command due to low pressure, when the air pressure rises above 9 bar.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_08.02.Zefiro-Europe.CONCEPT.211	When a train shut down request is issued (BC1 opening procedure started from driver or automatically due train unmanned for 10 minutes) and the minimum value in the last 24 hours of outside air temperature was below 5 °C for at least 1 hour, the TCMS shall set the “partial freeze drain” command, if not already requested by IDU, and send it to the toilet system via Ethernet.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_08.02.Zefiro-Europe.CONCEPT.212	In order to complete the shut down procedure (i.e opening the main battery contactor), the TCMS shall receive the “Partial freeze drain done”=2,3,4 feedback signal from all the toilets if “partial freeze drain” procedure has been previously commanded to start. In any case the complete shut down will be completed after T.B.D minutes.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_08.02.Zefiro-Europe.CONCEPT.214	It shall be possible to set and reset the “partial freeze drain” command of all the toilets from the IDUs.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_08.02.Zefiro-Europe.CONCEPT.215	When the “partial freeze drain” command is set the toilet system shall start the “partial freeze drain” routine (empting of the complete piping of the fresh water system and the toilet itself).	Derived Requirement	4S_08.03.-.Toilet Systems		N/A

2F_08.02.Zefiro-Europe.CONCEPT.216	When the “partial freeze drain” routine is positively completed the toilet system shall send a feedback to the TCMS via Ethernet.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.217	When the “partial freeze drain” routine fails the toilet system shall send a feedback to the TCMS	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.218	TCMS shall store an event (Warning) when the “partial freeze drain” routine fails	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
~~2F_08.02.Zefiro-Europe.CONCEPT.220~~	~~If the toilet normally powered and detects a temperature below a preset value [TBD], the “partial freeze drain” routine shall be performed. The toilet system shall come back to the normal operation when the temperature rises above a preset value [TBD].~~	~~Derived Requirement~~	~~4S_08.03.-.Toilet Systems~~		~~N/A~~
2F_08.02.Zefiro-Europe.CONCEPT.219	The toilet system shall remain in the partial freeze drain mode as long as the “partial freeze drain” command coming from the TCMS is active.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.HR03	The toilet system shall activate the partial freeze drainage when it is ordered via the MMI interface of the toilet control unit.	New Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.HR04	It shall be possible to set and reset the “partial freeze drain” command via the MMI interface of the toilet control unit.	New Requirement	4S_08.03.-.Toilet Systems		N/A

3.2.6 Toilet fire alarm

3.2.6.1 Design description

See Fire protection and extinction system SFRS.

3.2.6.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.96
Description	Smoke/fire shall be detected in toilets.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.02.Zefiro-Europe.CONCEPT.HR05	In case of toilet’s fire alarm signal received, the TCMS sends a command via ethernet to the toilet’s control unit in order to activate the alarm lamp and the buzzer installed in the toilet.	New Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.HR06	The toilet system shall remain in the fire alarm mode as long as the “fire alarm” command coming from the TCMS is active.	New Requirement	4S_08.03.-.Toilet Systems		N/A

3.3 Supervision

3.3.1 TCMS Communication

3.3.1.1 Design description

In order to increase its availability, the toilet system is able to work also in case of IP network failure.
Also in case of that failure a minimum level of communication between toilet and TCMS is assured by the “Out of order signal” line. It is a hardwired line that is at battery level voltage when the toilet system is correctly working.

3.3.1.2 Requirements

ID	2F_08.02.Zefiro-Europe.TRS.78
Description	The Toilet system shall continue to work in case there is a communication failure with TCMS
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.02.Zefiro-Europe.CONCEPT.65	The toilet system shall continue to work also in case of a failure on the IP communication.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.124	The toilet system shall activate a "Out of order" signal at high level (+batt) when the toilet is powered and working correctly.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.02.Zefiro-Europe.CONCEPT.125	TCMS shall store an event (Warning) in case of deactivation of the toilet "Out of order" signal.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_08.02.Zefiro-Europe.CONCEPT.126	System status of the toilet shall be shown on a suitable IDU page	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_08.02.Zefiro-Europe.CONCEPT.226	The Toilet system "Out of order" signal shall be sent to TCMS using a dedicated harwired line.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_08.02.Zefiro-Europe.CONCEPT.28			

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_08.02.Zefiro-Europe.CONCEPT.29	UNI toilet Alarm (UNI toilet only)	L	2F_08.02.Zefiro-Europe.CONCEPT.58
2F_08.02.Zefiro-Europe.CONCEPT.127	"Out of order" signal	L	2F_08.02.Zefiro-Europe.CONCEPT.124

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_08.02.Zefiro-Europe.CONCEPT.30	UNI toilet Alarm (UNI toilet only))	L	2F_08.02.Zefiro-Europe.CONCEPT.58
2F_08.02.Zefiro-Europe.CONCEPT.128	"Out of order" signal	L	2F_08.02.Zefiro-Europe.CONCEPT.124

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_08.02.Zefiro-Europe.CONCEPT.31			

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_08.02.Zefiro-Europe.CONCEPT.32	IFWTankContent1 (FWT level 0, ¼, ½, ¾, 1)	A	2F_08.02.Zefiro-Europe.CONCEPT.51
2F_08.02.Zefiro-Europe.CONCEPT.129	IFWTankEmpty1 (FWT completely Empty)	L	2F_08.02.Zefiro-Europe.CONCEPT.111 2F_08.02.Zefiro-Europe.CONCEPT.194
2F_08.02.Zefiro-Europe.CONCEPT.130	IWWTankFull (WWT completely Full)	L	2F_08.02.Zefiro-Europe.CONCEPT.52
2F_08.02.Zefiro-Europe.CONCEPT.131	IWWTankFull (WWT 80% level)	L	2F_08.02.Zefiro-Europe.CONCEPT.52
2F_08.02.Zefiro-Europe.CONCEPT.132	IWWTankFull (WWT 50% level)	L	2F_08.02.Zefiro-Europe.CONCEPT.52
2F_08.02.Zefiro-Europe.CONCEPT.133	IFreezDrainDone (Partial freeze drain done)	L	2F_08.02.Zefiro-Europe.CONCEPT.216
2F_08.02.Zefiro-Europe.CONCEPT.134	IFreezDrainFailure	L	2F_08.02.Zefiro-Europe.CONCEPT.217

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_08.02.Zefiro-Europe.CONCEPT.33	CFreezeDrain1 (complete freeze drainage request)	L	2F_08.02.Zefiro-Europe.CONCEPT.122
2F_08.02.Zefiro-Europe.CONCEPT.224	CFreezeDrain2 (partial freeze drainage request)	L	2F_08.02.Zefiro-Europe.CONCEPT.214
2F_08.02.Zefiro-Europe.CONCEPT.HR07	CFireAlarm	L	2F_08.02.Zefiro-Europe.CONCEPT.HR05

5 References

[1] EC01P006860B Toilet TRD

[2] EC010019276 Toilet VZI-6 TRD

[3] FA020025419 Fire protection and extinction system - SFRS